On the Failure of Finite-Width Extraction for ALCI_RCC5

A counterexample to the $\mathsf{FW}(C, N)$ conjecture
and its implications for the contextual tableau approach

Claude (Anthropic)
Prompted by Michael Wessel

March 2026

Disclaimer. This document was generated by Claude (Anthropic), prompted by Michael Wessel. The results have not been independently verified. We invite scrutiny and corrections.

Abstract

A recent contextual tableau calculus for ALCI_RCC5 [3] reduces the decidability question to the *finite-width extraction* conjecture $\mathsf{FW}(C, N)$: every satisfiable concept C admits, at some computable width bound $N(C)$, a finite closed family of contextual local states. We show that $\mathsf{FW}(C, N)$ is **false** for every N when C is the concept $(\exists \mathsf{PP}.\top) \sqcap (\forall \mathsf{PP}.\exists \mathsf{PP}.\top)$, which is satisfiable only in infinite models. The argument is short: the PP-relation in any local state is a finite strict partial order, hence has maximal elements, whose existential PP-demands cannot be met by the recentering mechanism. The counterexample shows that the contextual tableau framework, as formulated, cannot serve as a complete decision procedure for ALCI_RCC5. The decidability of ALCI_RCC5 remains open.

1 Introduction

The description logic ALCI_RCC5 extends $\mathcal{ALCI}$ with a role box derived from the RCC5 composition table. Wessel [1] introduced the logic and left decidability open, noting that ALCI_RCC5 lacks both the finite model property and the tree model property. In particular, the concept

$$C_\infty \;=\; (\exists \mathsf{PP}.\top) \sqcap (\forall \mathsf{PP}.\exists \mathsf{PP}.\top) \tag{1}$$

is satisfiable only in infinite models [1, 2].

A 2026 discussion draft [2] proposed a quasimodel approach but identified an *extension gap* in the completeness direction. A companion note [3] proposed a *contextual tableau calculus* that avoids the extension gap by re-designing the shape of a tableau node: each node is a finite contextual local state—a bounded-width atomic RCC5 network of remembered items with witness assignments and recentering maps. The note proves full *soundness* (every open tableau graph unfolds into a genuine model) but isolates completeness as an open conjecture:

> $\mathsf{FW}(C, N)$: *Every model of C contains, after quotienting by a finite descriptor equivalence, a closed family of local states of width at most N.*

If $\mathsf{FW}(C, N(C))$ holds for some computable bound $N(C)$, then combining it with soundness yields decidability [3, Proposition 7.1].

The purpose of this note is to show that $\mathsf{FW}(C_\infty, N)$ fails for **every** N. The argument is elementary and rests on a single observation: PP is a strict partial order, and finite strict partial orders have maximal elements.

2 Recollection of the framework

We recall only what is needed for the counterexample; see [3] for full definitions.

2.1 Local states and closed families

Fix a concept C in negation normal form and let $\mathrm{cl}(C)$ denote its closure under subformulas.

Definition 2.1 (local state, [3, Definition 3.1])**.** A *local state* of width at most N is a tuple $M = (I_M, 0_M, \mathrm{tp}_M, \rho_M, w_M)$ where:

(a) I_M is a finite nonempty set of *items* with $|I_M| \leq N$ and a distinguished *center* $0_M \in I_M$.

(b) $\mathrm{tp}_M : I_M \to 2^{\mathrm{cl}(C)}$ assigns Hintikka types.

(c) $\rho_M : I_M \times I_M \to \mathsf{R}$ is a total, path-consistent atomic RCC5 labeling with $\rho_M(i, i) = \mathsf{EQ}$ and $\rho_M(j, i) = \mathrm{inv}(\rho_M(i, j))$.

(d) For every $\exists R.D \in \mathrm{tp}_M(0_M)$, a designated witness $w_M(\exists R.D) \in I_M$ with $\rho_M(0_M, w_M(\exists R.D)) = R$ and $D \in \mathrm{tp}_M(w_M(\exists R.D))$.

The state satisfies universal propagation (L1), transitive propagation (L2), and EQ-synchrony (L3).

Definition 2.2 (closed family / open tableau graph, [3, Definitions 4.1, 5.1])**.** A *closed family of width* N for C consists of:

- a finite set $\mathcal{F}$ of saturated local states of width $\leq N$, with a distinguished root state containing C in its center type;
- for each $M \in \mathcal{F}$ and each $i \in I_M$, a successor state $\mathrm{Succ}(M, i) \in \mathcal{F}$;
- for each such edge, a total *recentering map* $u_{M,i} : I_{\mathrm{Succ}(M,i)} \to I_M$ satisfying the axioms:

(R1) $u_{M,i}(0_{\mathrm{Succ}(M,i)}) = i$;

(R2) $\mathrm{tp}_{\mathrm{Succ}(M,i)}(a) = \mathrm{tp}_M(u_{M,i}(a))$ for all a;

(R3) $\rho_{\mathrm{Succ}(M,i)}(a, b) = \rho_M(u_{M,i}(a), u_{M,i}(b))$ for all a, b;

(R4) $\mathrm{Succ}(\mathrm{Succ}(M, i), a) = \mathrm{Succ}(M, u_{M,i}(a))$ for all a.

The key result of [3] is:

Theorem 2.3 (Soundness, [3, Corollary 6.6])**.** *If a closed family of width N for C exists, then C is satisfiable in* $\mathsf{ALCI_RCC5}$ *under strong* EQ *semantics.* □

The conditional completeness says: if $\mathsf{FW}(C, N)$ holds, then C is satisfiable *if and only if* a closed family of width N exists.

3 The counterexample

3.1 The concept and its models

Consider

$$C_\infty = (\exists \mathsf{PP}.\top) \sqcap (\forall \mathsf{PP}.\exists \mathsf{PP}.\top).$$

As observed by Wessel [1] and Claude [2, Remark 2.5], C_∞ is satisfiable only in infinite models. A canonical model is the PP-chain $\mathcal{I}$ with domain $\Delta^{\mathcal{I}} = \{d_0, d_1, d_2, \dots\}$ and

$$\rho(d_i, d_j) = \begin{cases} \mathsf{PP} & \text{if } i < j, \\ \mathsf{EQ} & \text{if } i = j, \\ \mathsf{PPI} & \text{if } i > j. \end{cases}$$

Every d_n satisfies $\exists \mathsf{PP}.\top$ (witnessed by d_{n+1}) and $\forall \mathsf{PP}.\exists \mathsf{PP}.\top$ (every PP-successor d_m with $m > n$ has its own PP-successor d_{m+1}).

3.2 The type forced by transitive propagation

Let τ be the Hintikka type that contains $\exists \mathsf{PP}.\top$ and $\forall \mathsf{PP}.\exists \mathsf{PP}.\top$.

Lemma 3.1. *In any closed family for C_∞, every item i of every state M that satisfies $\rho_M(0_M, i) = \mathsf{PP}$ must have $\exists \mathsf{PP}.\top \in \mathrm{tp}_M(i)$ and $\forall \mathsf{PP}.\exists \mathsf{PP}.\top \in \mathrm{tp}_M(i)$.*

Proof. The center 0_M has $\forall \mathsf{PP}.\exists \mathsf{PP}.\top \in \mathrm{tp}_M(0_M)$ (since $C_\infty \in \mathrm{tp}_M(0_M)$ for the root state, and for successor states the type is inherited via (R2)).

By universal propagation (L1): since $\forall \mathsf{PP}.\exists \mathsf{PP}.\top \in \mathrm{tp}_M(0_M)$ and $\rho_M(0_M, i) = \mathsf{PP}$, we get $\exists \mathsf{PP}.\top \in \mathrm{tp}_M(i)$.

By transitive propagation (L2): since PP is transitive and $\forall \mathsf{PP}.\exists \mathsf{PP}.\top \in \mathrm{tp}_M(0_M)$ with $\rho_M(0_M, i) = \mathsf{PP}$, we get $\forall \mathsf{PP}.\exists \mathsf{PP}.\top \in \mathrm{tp}_M(i)$. □

In the successor state $\mathrm{Succ}(M, i)$, item i becomes the center. By (R2), the center of $\mathrm{Succ}(M, i)$ has the same type as item i in M. If that type contains $\exists \mathsf{PP}.\top$ and $\forall \mathsf{PP}.\exists \mathsf{PP}.\top$, then $\mathrm{Succ}(M, i)$ is again a state whose center has these formulas, and the lemma applies to its items as well.

Hence: in any closed family for C_∞, every state reachable from the root has a center containing $\exists \mathsf{PP}.\top$ and $\forall \mathsf{PP}.\exists \mathsf{PP}.\top$. Every PP-witness in such a state also has these formulas in its type.

3.3 The strict partial order obstruction

Lemma 3.2. *In any local state M, the relation PP on I_M is a finite strict partial order (irreflexive, asymmetric, transitive).*

Proof. Irreflexivity: $\rho_M(i, i) = \mathsf{EQ} \neq \mathsf{PP}$ for all i.

Asymmetry: if $\rho_M(i, j) = \mathsf{PP}$ then $\rho_M(j, i) = \mathrm{inv}(\mathsf{PP}) = \mathsf{PPI} \neq \mathsf{PP}$.

Transitivity: if $\rho_M(i, j) = \mathsf{PP}$ and $\rho_M(j, k) = \mathsf{PP}$, then by the RCC5 composition table, $\rho_M(i, k) \in \mathrm{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$, so $\rho_M(i, k) = \mathsf{PP}$. □

Lemma 3.3. *Every finite strict partial order has at least one maximal element.*

Proof. Standard: if every element had a strict successor, iterating from any element would produce an infinite chain, contradicting finiteness. □

3.4 The failure of recentering

We now prove the main result.

Theorem 3.4 (Failure of $\mathsf{FW}(C_\infty, N)$)**.** *For every $N \geq 1$, there is no closed family of width N for C_∞.*

Proof. Suppose for contradiction that $\mathcal{F}$ is a closed family of width N for C_∞ with root state M_*.

Step 1: All reachable states have uniform type structure. By Lemma 3.1 and induction on the successor chain, every state M reachable from M_* has $\exists\mathsf{PP}.\top \in \mathrm{tp}_M(0_M)$. The designated PP-witness for this formula is some item $w \in I_M$ with $\rho_M(0_M, w) = \mathsf{PP}$. By Lemma 3.1, $\exists\mathsf{PP}.\top \in \mathrm{tp}_M(w)$ and $\forall\mathsf{PP}.\exists\mathsf{PP}.\top \in \mathrm{tp}_M(w)$.

Moreover, in the successor state $\mathrm{Succ}(M, w)$: the center has the same type as w in M, which includes $\exists\mathsf{PP}.\top$ and $\forall\mathsf{PP}.\exists\mathsf{PP}.\top$. So the analysis repeats. By induction, *every state reachable along witness-successor chains has a center with $\exists\mathsf{PP}.\top$ in its type.*

Step 2: Every item of a reachable state needs a PP-successor. Actually, we need to be precise. Not every item needs $\exists\mathsf{PP}.\top$ in its type—only items that are PP from the center. But for the argument it suffices to focus on a single chain.

Start from the root state M_*. Its center satisfies $\exists\mathsf{PP}.\top$, so it has a PP-witness $w_1 \in I_{M_*}$. Consider $M_1 = \mathrm{Succ}(M_*, w_1)$. Its center has type containing $\exists\mathsf{PP}.\top$ (by (R2) and Lemma 3.1), so it has a PP-witness $w_2 \in I_{M_1}$. By (R1) and (R3), the recentering map $u : I_{M_1} \to I_{M_*}$ must satisfy:

$$u(0_{M_1}) = w_1, \tag{2}$$

$$\rho_{M_1}(0_{M_1}, w_2) = \mathsf{PP}, \tag{3}$$

$$\rho_{M_*}(u(0_{M_1}), u(w_2)) = \rho_{M_1}(0_{M_1}, w_2) = \mathsf{PP}. \tag{4}$$

Combining (2) and (4):

$$\rho_{M_*}(w_1, u(w_2)) = \mathsf{PP}. \tag{5}$$

So $u(w_2)$ must be an item in I_{M_*} that is PP from w_1.

Step 3: Iterating produces a PP-chain in I_{M_*}. Set $a_0 = 0_{M_*}$ and $a_1 = w_1$. By (5), $u(w_2)$ is some item $a_2 \in I_{M_*}$ with $\rho_{M_*}(a_1, a_2) = \mathsf{PP}$.

Now consider $M_2 = \mathrm{Succ}(M_1, w_2) = \mathrm{Succ}(M_*, u(w_2)) = \mathrm{Succ}(M_*, a_2)$ (by (R4)). The center of M_2 has type containing $\exists\mathsf{PP}.\top$ (same argument), so it has a PP-witness w_3. The recentering $u' : I_{M_2} \to I_{M_*}$ satisfies $u'(0_{M_2}) = a_2$ and $\rho_{M_*}(a_2, u'(w_3)) = \mathsf{PP}$. Set $a_3 = u'(w_3)$.

Continuing, we obtain a sequence $a_0, a_1, a_2, a_3, \ldots$ of items in I_{M_*} with $\rho_{M_*}(a_k, a_{k+1}) = \mathsf{PP}$ for all $k \geq 0$.

Step 4: Contradiction. By Lemma 3.2, PP is a strict partial order on I_{M_*}. In particular, PP is antisymmetric and transitive. For $j < k$ we have $\rho_{M_*}(a_j, a_k) = \mathsf{PP}$ (by transitivity of the chain), hence $a_j \neq a_k$. Therefore $a_0, a_1, a_2, \ldots$ are pairwise distinct items in I_{M_*}.

But $|I_{M_*}| \leq N$, a finite bound. An infinite sequence of pairwise distinct items in a set of size N is impossible. Contradiction. □

Remark 3.5. The proof does *not* claim that C_∞ is unsatisfiable. It is satisfiable—in infinite models. The theorem says only that no *finite closed family of finite-width local states* can capture C_∞. Soundness of the tableau ([3, Corollary 6.6]) is untouched; it is completeness that fails.

Remark 3.6. The argument generalizes immediately to any concept C such that (i) C is satisfiable only in infinite models, and (ii) the infinite model is forced by an unending PP-chain (or PPI-chain) where every element in the chain must have a PP-successor (or PPI-successor) of the same character. The root cause is the transitivity of PP and PPI.

4 Why the obstruction arises and why it is fundamental

4.1 The role of strict partial orders

The recentering axiom (R3) requires that RCC5 relations in a successor state are faithfully reflected in the parent state. When the relation in question is PP, this means: if the successor state contains a PP-edge, the parent state must contain a corresponding PP-edge between the recentered images.

Because PP is transitive and antisymmetric, iterating this requirement along a chain of successor states produces a growing PP-chain *inside a single parent state*. Finiteness of the state forces the chain to terminate, but the ∃PP demands force it to continue. This is the contradiction.

4.2 Comparison with the tree model property

In standard $\mathcal{ALCI}$ (without RCC constraints), every satisfiable concept has a tree model. Tree models can be made regular (finitely many subtree-types) because there is no interaction between branches: the tree structure provides natural "blocking."

In ALCI_RCC5, the model is a complete graph. The PP-relation threads through the entire graph and enforces a global partial order. There is no way to "block" a PP-chain because a blocked node would need a PP-successor that is *strictly deeper* in the partial order—reusing an ancestor is impossible (it would reverse PP). This is the fundamental obstacle that distinguishes ALCI_RCC5 from standard description logics.

4.3 Comparison with the quasimodel approach

The quasimodel method of [2] has a complementary gap. There, *soundness* (satisfiable concept ⇒ quasimodel) is proven, but *completeness* (quasimodel ⇒ model) has a gap: the Henkin construction may fail for abstract quasimodels.

The contextual tableau of [3] has the opposite gap: *soundness* (closed family ⇒ model) is proven, but *completeness* (satisfiable concept ⇒ closed family) fails outright for concepts like C_∞.

	Quasimodel [2]	Contextual tableau [3]
Satisfiable $\Rightarrow$ representation	Proven	**False** for C_∞
Representation $\Rightarrow$ model	**Gap**	Proven

Neither approach, as formulated, yields a decision procedure.

5 Possible directions forward

The failure of $\mathsf{FW}(C, N)$ does not settle the decidability question. It eliminates one specific proof strategy (fixed-width closed families) but leaves several alternatives open.

1. **Variable-width or infinite-width representations.** The finite-width constraint is the source of the obstruction. One could allow the width of local states to depend on the *unfolding depth* rather than being globally bounded. The challenge is ensuring that the resulting search space remains finite (or decidable).
2. **Automata-theoretic approaches.** Two-way alternating tree automata handle $\mathcal{ALCI}$ decidability elegantly. Adapting them to the complete-graph setting of $\mathsf{ALCI_RCC5}$ is non-trivial, but the regular structure of the infinite PP-chain (every node has the same type) suggests that some form of ω-regularity or Büchi acceptance might capture the required infinite patterns.
3. **Reduction to constraint satisfaction.** The quasimodel approach of [2] reduces the problem to checking whether abstract quasimodels (which can be finitely enumerated) yield genuine models. Closing the extension gap in that approach—showing that every abstract quasimodel satisfying (Q1)–(Q3) is realizable—would still yield decidability. The present counterexample does not affect that direction.
4. **Undecidability.** It is also possible that $\mathsf{ALCI_RCC5}$ is undecidable. The PP-chain phenomenon and the absence of finite model or tree model properties are suggestive, though not conclusive. A reduction from a known undecidable problem (e.g., the halting problem or the tiling problem) to $\mathsf{ALCI_RCC5}$ satisfiability would settle the question negatively.

6 The counterexample for RCC8

The same argument applies to $\mathsf{ALCI_RCC8}$. In RCC8, the role NTPP (non-tangential proper part) is a strict partial order, with composition $\mathsf{NTPP} \circ \mathsf{NTPP} = \{\mathsf{NTPP}\}$. The concept

$$C'_\infty = (\exists \mathsf{NTPP}.\top) \sqcap (\forall \mathsf{NTPP}.\exists \mathsf{NTPP}.\top)$$

forces an infinite NTPP-chain, and the strict partial order argument of Theorem 3.4 applies verbatim. So $\mathsf{FW}(C'_\infty, N)$ fails for every N in the RCC8 setting as well.

7 Conclusion

We have shown that the finite-width extraction conjecture $\mathsf{FW}(C, N)$ is false for the satisfiable $\mathsf{ALCI_RCC5}$ concept $C_\infty = (\exists \mathsf{PP}.\top) \sqcap (\forall \mathsf{PP}.\exists \mathsf{PP}.\top)$. The proof is elementary: transitivity of PP

and the recentering axiom (R3) together force an infinite PP-chain inside any candidate local state, contradicting the finite width bound.

This closes the completeness direction of the contextual tableau approach of [3] negatively. Combined with the extension gap in the quasimodel approach of [2], the decidability of ALCI_RCC5 (and ALCI_RCC8) remains an open problem. Both approaches have been shown to have genuine, non-trivial gaps that are not mere artifacts of incomplete proofs but reflect fundamental structural features of the logic.

The key structural feature is the combination of (i) the transitivity of PP (creating strict partial orders), (ii) the universal propagation of ∀PP along PP-chains, and (iii) the requirement that models be complete graphs. Any successful decidability proof—or undecidability reduction—for ALCI_RCC5 will need to engage with this combination directly.

References

[1] Michael Wessel. *Qualitative Spatial Reasoning with the ALCIRCC Family – First Results and Unanswered Questions*. Technical Report FBI-HH-M-324/03, University of Hamburg, 2002/2003.

[2] Claude. *On the Decidability of ALCIRCC5 and ALCIRCC8: Quasimodels Meet the Patchwork Property*. Discussion draft, 2026.

[3] GPT-5.4 Pro. *A Contextual Tableau Calculus for* ALCI_RCC5. Discussion draft, 2026.

[4] Jochen Renz and Bernhard Nebel. On the complexity of qualitative spatial reasoning: A maximal tractable fragment of the Region Connection Calculus. *Artificial Intelligence*, 108(1–2):69–123, 1999.

[5] Jochen Renz. Maximal tractable fragments of the Region Connection Calculus: A complete analysis. In *Proc. IJCAI*, pages 448–454, 1999.